

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

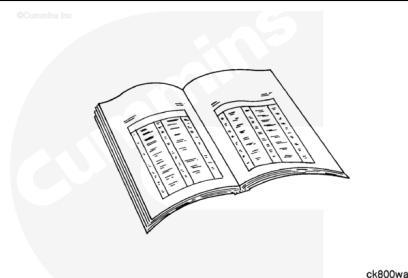
⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapor, ingestion, and prolonged contact with used engine oil. If not reused dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

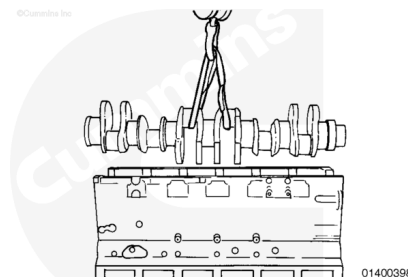
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>).
- Remove the oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>).
- Remove the connecting rod caps. Refer to Procedure 001-054 in Section 1 (</qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html>).
- Remove the main bearing caps and bearings. Refer to Procedure 001-006 in Section 1 (</qs3/pubsys2/xml/en/procedures/18/18-001-006.html>).
- Remove the thrust bearings. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-007.html>)



Remove

Support the weight of the crankshaft with a hoist or lifting device.

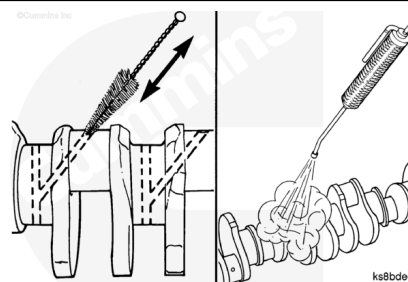
Remove the crankshaft.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Clean the crankshaft with solvent and dry with compressed air.

Use crocus cloth or a Scotch-Brite™ pad to remove discoloration or light scratches from the machined surfaces.



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When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

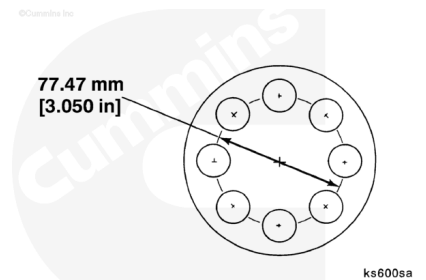
⚠ CAUTION ⚠

Do not use a thread chaser to clean the capscrew threads in the crankshaft. Severe engine damage can result.

To clean the ROLLED threads, flush with solvent, and dry with compressed air.

If additional cleaning is required, brush with a nylon bristle brush.

Place tape over the threaded capscrew holes.

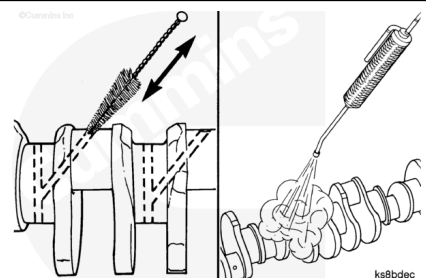


⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

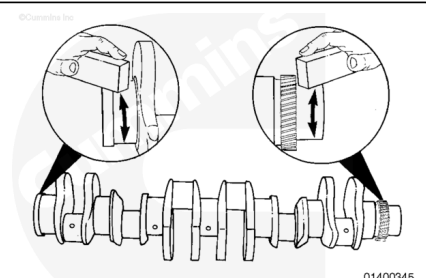
Use a bristle brush and solvent to clean all of the crankshaft oil drillings.

Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.



⚠ WARNING ⚠

When using solvents, acids or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



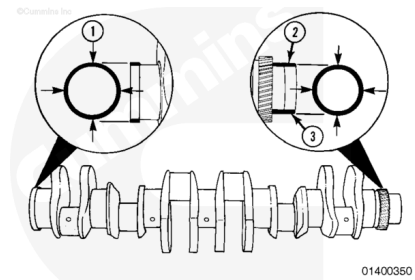
New crankshafts are coated with a heavy preservative. Use solvent to thoroughly remove the coating. Brush or flush the packing debris from the oil drillings before installing the crankshaft into the engine.

Use a honing stone to polish the outside diameter of the front and rear oil seal locations, the flywheel mounting location, and the vibration damper location. Remove all small scratches and grooves.

Use a light preservative oil to prevent rust during engine rebuild. If the crankshaft is **not** going to be installed immediately, use a heavy preservative oil.

Measure the outside diameter at the locations shown.

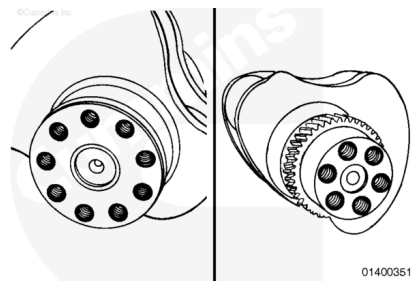
Crankshaft Outside Diameter			
	mm		in
Location (1)	152.35	MIN	5.998
	152.40	MAX	6.000
Location (2)	111.07	MIN	4.373
	111.13	MAX	4.375
Location (3)	110.74	MIN	4.360
	110.77	MAX	4.361



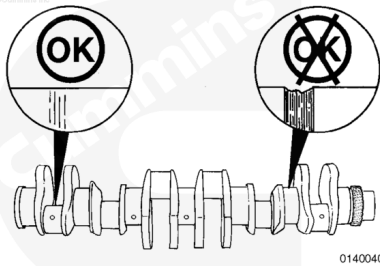
⚠ CAUTION ⚠

Do not chase threads on the crankshaft. Severe engine damage can occur.

Check the threads for damage at both ends of the crankshaft. If necessary, refer to the Alternative Repair Manual, Bulletin 3379035, for repair instructions.



Check the main bearing journals and rod bearing journals for damage or excessive wear. Minor scratches are acceptable.



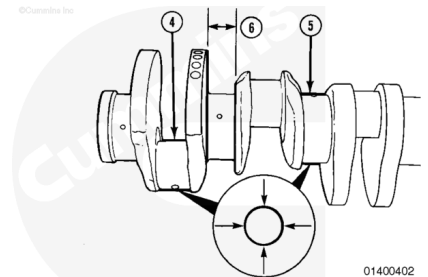
Measure the outside diameter.

Rod Bearing Journal Outside Diameter (4)

mm		in
101.549	MIN	3.998
101.600	MAX	4.000

Main Bearing Journal Outside Diameter (5)

mm		in
139.649	MIN	5.498
139.700	MAX	5.5000



Measure the thrust distance between the thrust faces on the number 6 main bearing journal.

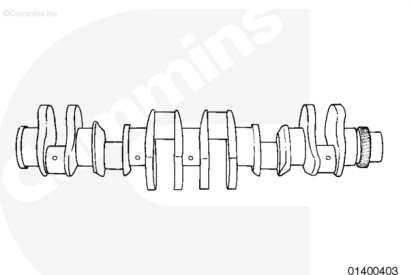
Thrust Distance (6)

mm		in
60.30	MIN	2.374
60.35	MAX	2.376

The crankshaft can be ground undersize if the outside diameter is **not** within specifications. **Always** grind all the journals when one is **not** within specifications.

Oversize rod bearings and main bearings are available. Refer to the Alternate Repair Manual, Bulletin 3379035, for grinding specifications and instructions.

Oversize thrust bearings are available if the thrust distance is **not** within specifications. A 0.25 mm [0.010 in] and a 0.51 mm [0.020 in] oversize thrust bearing are available. If the crankshaft **must** be machined to use an oversize thrust bearing, refer to the Alternate Repair Manual, Bulletin 3379035, for instructions.

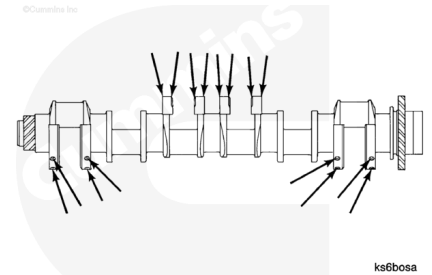


Use a light preservative oil. Lubricate the crankshaft to prevent rust. If the crankshaft is **not** going to be used immediately, use a heavy preservative oil.

Note : Only on engines with removable counterweights:

⚠ CAUTION ⚠

The capscrews must be tight or the counterweight can fall, causing severe damage to the engine. Any loose capscrews must be replaced with new capscrews.



⚠ CAUTION ⚠

Capscrews cannot be used again and must be replaced with new capscrews.

⚠ CAUTION ⚠

Check new capscrews for any visible damage or wear. Discard the capscrew if there is any visible damage or wear to the thread, neck, or capscrew head.

Use a torque wrench to check the torque of the counterweight capscrews. Each counterweight contains two capscrews.

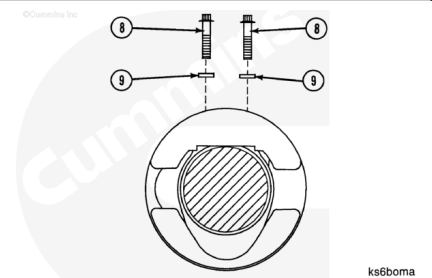
Torque Value: 142 n·m [105 ft-lb]

Mark any counterweights that have a loose capscrew.

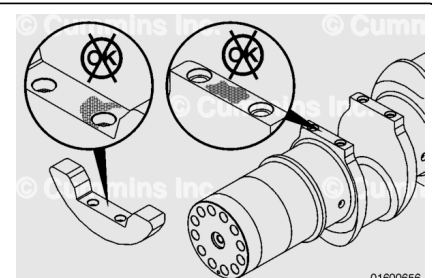
Note : The following step applies only to the counterweights that contain a loose capscrew.

Remove the capscrews (8) and washers (9).

Remove the counterweight.



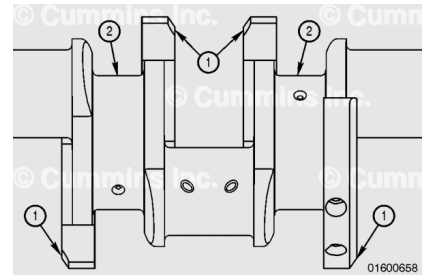
Check the counterweight and crankshaft for fretting or damage between the mating surfaces. If the counterweight is damaged, the counterweight **must** be replaced and the crankshaft **must** be balanced. If the crankshaft is damaged, the crankshaft **must** be replaced or the counterweight replaced and the crankshaft repaired.



Not more than 0.51 mm [0.020 in] can be removed from the crankshaft mounting pad to repair fretting.

⚠ CAUTION ⚠

If the counterweight is removed, it must be installed with the part number toward the nearest main bearing journal. If the counterweight is not installed correctly, the weight can strike the connecting rod, causing extensive damage to the engine.



Note : Make sure the counterweight is returned to its original location. The counterweights have the same number stamped on them as the web to which they are attached.

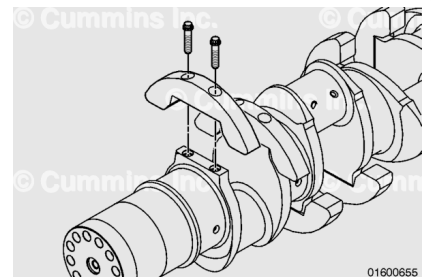
There **must not** be an interference between the crankshaft and counterweight and the sides of the mounting pad.

Lubricate the washers and capscrews with clean engine oil.

Install the washers (9) and capscrews (8).

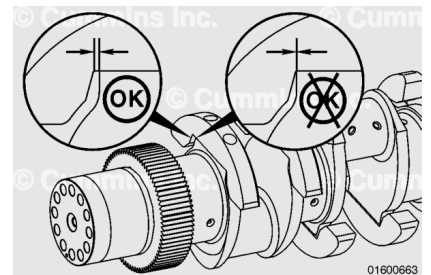
Tighten the capscrews.

Torque Value: 142 n·m [105 ft-lb]



With the capscrews tightened, visually inspect the counterweight for a interference condition with the crank.

There **must not** be an interference between the crankshaft and counterweight and the sides of the mounting pad.

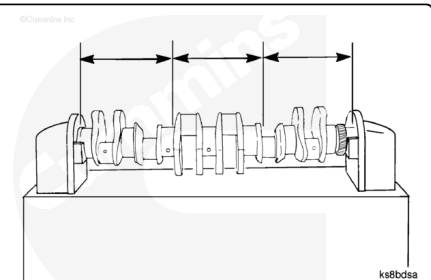


Magnetic Crack Inspect

Use a magnetic particle testing machine.

Perform the head shot and inspection method, then perform the coil shot and inspection method.

Adjust the testing machine to 1800 ampere direct current or rectified alternating current.

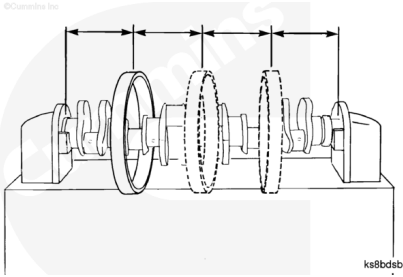


Use the continuous method. Wet **only** 1/3 of the crankshaft at a time.

Check the crankshaft for cracks.

Use the coil shot method. Use a coil that is a minimum of 514 mm [20.250 in] diameter.

Use the continuous method. Apply coil shot.



Amperage (Ampere Turns)	
Minimum	Maximum
4500 VDC or rectified alternating current	5000 VDC or rectified alternating current

Ampere turn is an electrical current of ampere flowing through the coil, multiplied by the number of turns in the coil.

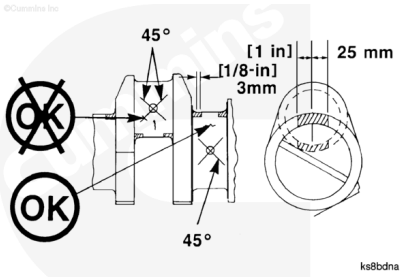
Check the crankshaft for cracks.

An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface can be seen with the use of the ultraviolet light.

Do **not** use the crankshaft if:

Limits of Acceptance - Open Indications

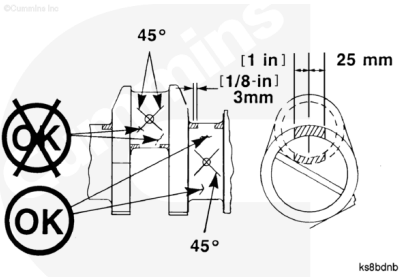
- There is an indication in the fillet or in the shaded area.
- There is an indication that passes through the 45 degree diagonal from the oil hole or goes into the oil hole chamfer.
- There is an indication that is longer than 6 mm [0.250 in].
- There are more than four indications on one journal.



Do **not** use the crankshaft if:

Limits of Acceptance - Indications Below the Surface

- There is an indication in the fillet or in the shaded area that is in a circumferential direction.
- There is an indication in a circumferential direction that is longer than 25.4 mm [1.0 in].
- There is an indication in an axial direction that is longer than 9.5 mm [0.375 in].



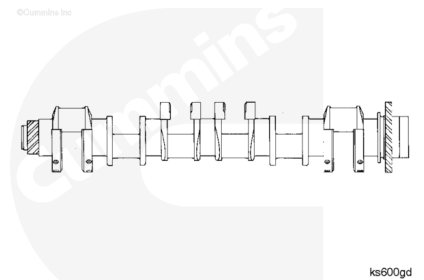
- There is an indication that is closer than 1.6 mm [0.063 in] to an oil hole chamfer.
- There is an indication that passes through the 45 degree diagonal from the oil hole.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ CAUTION ⚠

The crankshaft must be demagnetized completely and cleaned thoroughly. Small metal particles will cause engine damage.



Use steam to clean the crankshaft and oil drillings.

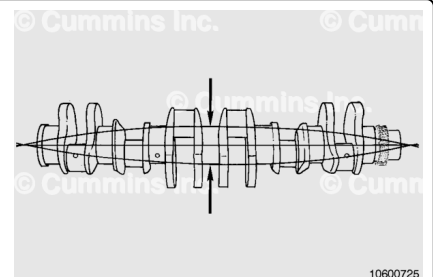
Use a light preservative oil to lubricate the crankshaft to prevent the formation of rust.

Use a heavy preservative oil if the crankshaft is **not** going to be installed immediately. Protect the part with a cover to prevent dirt from sticking to the oil.

Bend and Twist Inspect

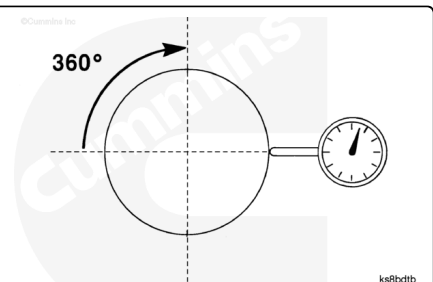
The crankshaft straightness is determined by the amount of total runout and the adjacent journal runout.

Total runout is defined as the total indicator runout measured at the middle bearing journal when the crankshaft is supported on the two end journals. Total runout is often referred to as the bend or full length alignment.



Journal runout is defined as the total indicator run out (total sweep of the needle) of the main bearing journals the crankshaft is rotated one complete revolution (360 degrees).

Adjacent journal runout is defined as the relationship of the total indicator runout of a main bearing journal as it is rotated on a common axis to the total indicator runout of an adjacent

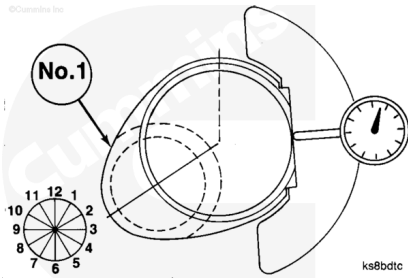


journal.

Adjacent runout is often referred to as step runout, bearing-to-bearing runout, or journal-to-journal runout.

The clock position is defined as the location of the journal at the highest total indicator runout point. Compare its angular relationship with the number one crankshaft pin, as viewed from the front of the crankshaft.

In the illustration, the crankshaft pin is at the 8 o'clock position. This is the clock position of the journal being measured.

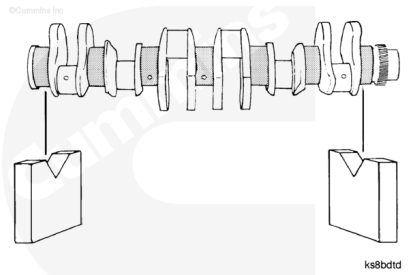


⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

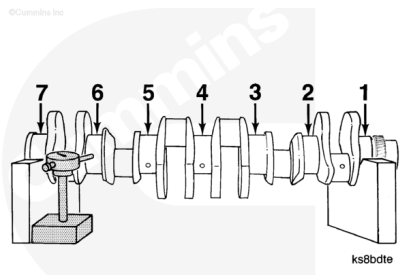
Place two vee-blocks on a flat surface.

Support the crankshaft on the vee-blocks at the two end bearing journals.



Position a dial indicator so the stem touches the center line of the main bearing journal.

The dial indicator **must** be positioned at the centerline of any journal that is measured (1) through (7).



Rotate the crankshaft and measure the total indicator runout at each bearing journal. Record the value and the clock position for each location.

ITEM	RUNOUT TIR [INCH]	CLOCK POSITION
JOURNAL STEP		
1	[0]	0
2	[0.0021]	12
3	[0.0030]	12
4	[0.0039]	1
5	[0.0025]	1
6	[0.0016]	2
7	[0]	0

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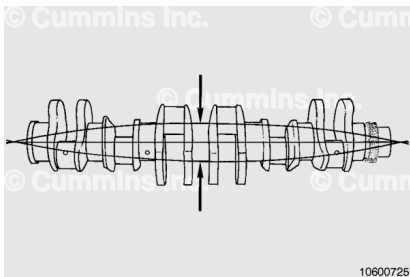
A fully fillet hardened crankshaft **must** be discarded if the total runout is **not** within specifications.

Position the dial indicator stem to the center main bearing journal.

Rotate the crankshaft a complete revolution.

Crankshaft Total Runout		
mm		in
0.230	MAX	0.009

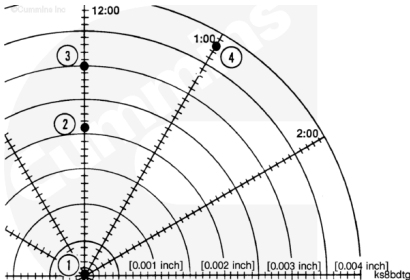
If the crankshaft is **not** within specifications, it **must** be replaced.



For each journal, plot the total indicator run out value at its clock position illustrated on the polar chart.

The end journals, supported by vee-blocks, **must** be plotted at the center of the chart.

The graphic illustrates the plot points.



Journal	Total Indicator Run Out	Clock Position
(1)	0	0
(2)	0.002	12
(3)	0.003	12
(4)	0.004	1

Draw a straight line between the plotted points. From journal number one to journal number two from journal number two to journal number three until all journals are plotted on the chart.

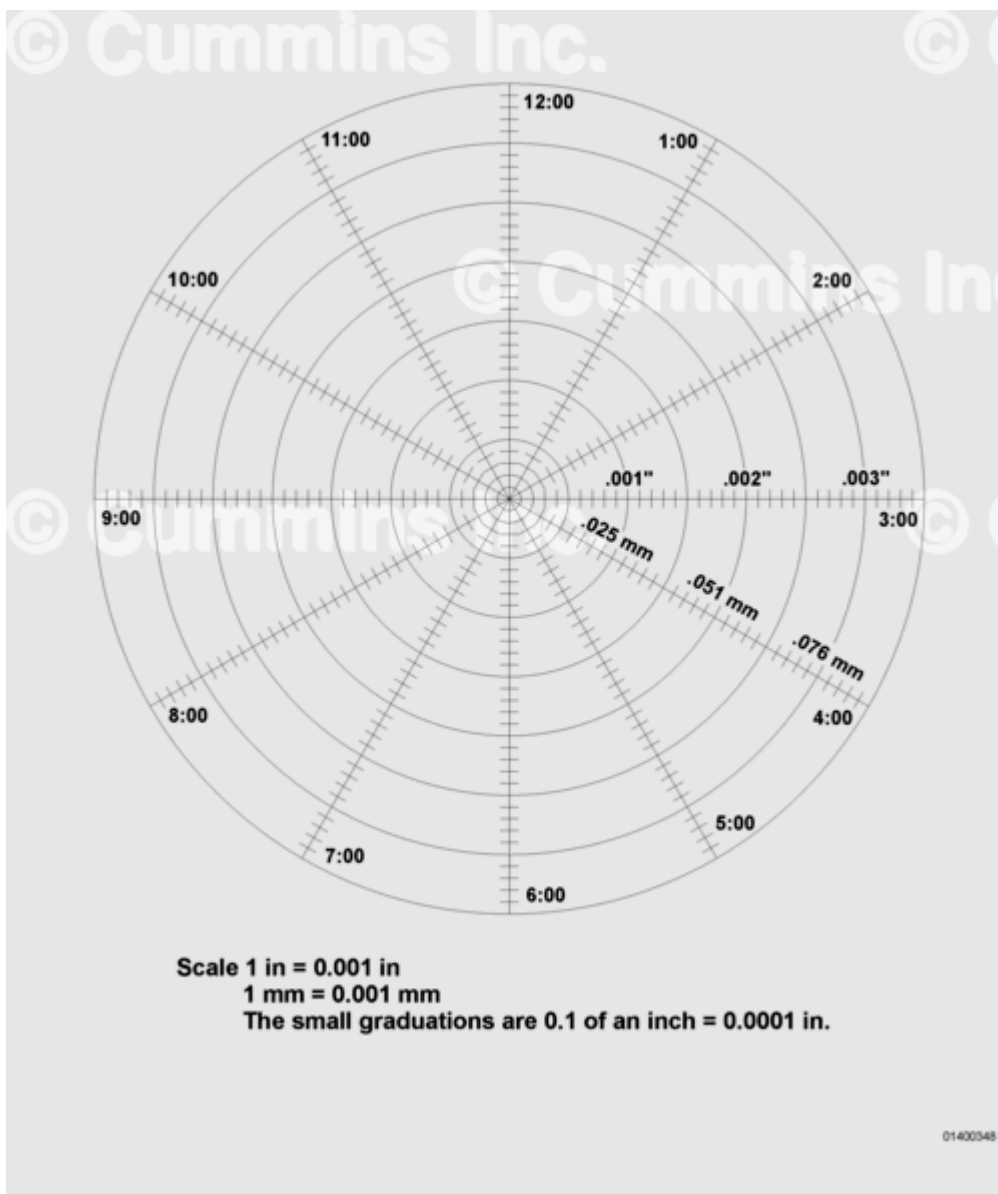
To determine the adjacent journal runout, measure the length of the line from each journal to its corresponding journal point.

In the above table journal number three and number four are 5.1 mm [2 in]. This represents a runout of 0.051 mm [0.002 in].

Record the adjacent journal runout for each main bearing journal.

The maximum adjacent journal runout is 0.08 mm [0.003 in].

If the crankshaft is **not** within specifications. It **must** be replaced.



Polar Chart

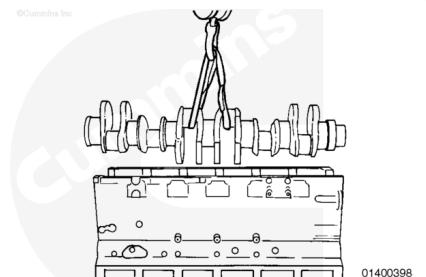
Polar chart

Install

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠



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Use a lifting strap that will not damage the crankshaft. Do not drop the crankshaft on the bearings.

Use a lint-free cloth to clean the crankshaft bearing journals.

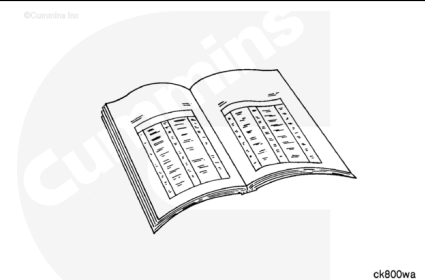
The end of the crankshaft with the smallest diameter **must** point toward the front of the block.

Install the crankshaft.

Finishing Steps

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.



ck800wa

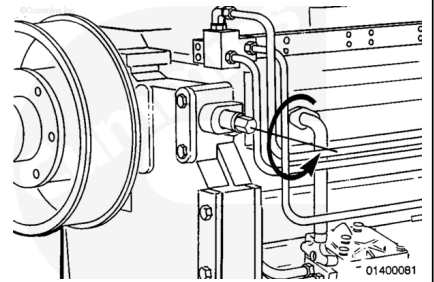
- Install the thrust bearings. Refer to Procedure 001-007 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-007.html>)
- Install the main bearings and caps. Refer to Procedure 001-006 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-006-tr.html>)
- Install the connecting rod caps. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/18/18-001-054-tr.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html>)
- Fill the engine with lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html>)

Rotation Check

To rotate the engine crankshaft, push in on the engine barring device and rotate **counterclockwise**.

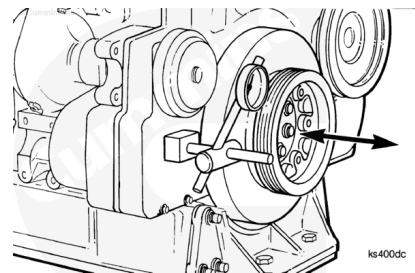
Rotate the crankshaft through two complete revolutions.

If the engine does **not** turn freely, the equipment can have a malfunction. Refer to the equipment manufacturer's instructions. The engine can have internal problems. Refer to correct procedure for inspection and replacement of internal engine components.



⚠ CAUTION ⚠

Extreme care must be used in prying against the viscous vibration damper. Sharp pry bars can damage the damper casing, resulting in a leak of the viscous vibration damper fluid and ultimate failure of the vibration damper.



Measure the crankshaft end clearance with a dial indicator.

Measure the end clearance.

Crankshaft End Clearance		
New Minimum	New Maximum	Worn Limit
0.10 mm [0.004 in]	0.43 mm [0.017 in]	0.56 mm [0.022 in]

The check can be made by attaching a dial indicator resting against the vibration damper or pulley while prying against the front cover and inner part of the pulley or vibration damper. End clearance **must** be in specification with the engine mounted in the unit and assembled to the transmission or converter.

If the clearance is **not** within specifications, contact a Cummins® Authorized Repair Location.

